## ARTI 404: Image Processing

# Image Denoising Using Spatial and Frequency Domain Methods

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Assignment

## Abstract

This report presents an image denoising experiment using two restoration approaches: a spatial-domain median filter and a frequency-domain Gaussian low-pass filter. Five noisy images were analyzed, denoised, tuned using multiple parameter settings, and evaluated using PSNR and SSIM. The clean images were used only as references for quantitative evaluation. The experiment shows that median filtering produced the strongest average performance for this dataset because most images contain impulse or salt-and-pepper noise.

## 1. Noise Identification

Each noisy image was visually inspected to determine the most likely noise type. The observations are summarized below.

| Image | Noise Type | Justification |
|---|---|---|
| camera | Gaussian noise | Fine grain appears across the whole image, which is typical of additive Gaussian noise. |
| coins | Salt-and-pepper / impulse noise | Many isolated bright and dark dots appear on coins and background, indicating impulse noise. |
| text | Salt-and-pepper noise | Random black and white pixels corrupt the paper and handwriting. |
| rice | Salt-and-pepper noise | Strong isolated bright/dark dots appear over the textured image. |
| astronaut | Colored salt-and-pepper / impulse noise | Random colored pixel corruption appears across RGB channels. |

## 2. Method Selection

### 2.1 Spatial-domain method: Median filter

The selected spatial-domain method is median filtering. Median filtering replaces each pixel with the median value in a local neighborhood. It is effective for impulse and salt-and-pepper noise because isolated corrupted pixels are removed without averaging them into the surrounding area. Kernel sizes 3x3, 5x5, 7x7, and 9x9 were tested.

### 2.2 Frequency-domain method: Gaussian Low-Pass Filter

The frequency-domain method applies a Fourier Transform to the image, shifts the zero-frequency component to the center, multiplies the spectrum by a Gaussian low-pass filter, then applies the inverse Fourier Transform. Cutoff values 10, 20, 40, and 80 were tested. Low-pass filtering reduces high-frequency noise but can also blur edges and fine details.

## 3. PSNR and SSIM Explanation

PSNR (Peak Signal-to-Noise Ratio) measures the pixel-level difference between the clean reference image and the denoised image. A higher PSNR means the denoised image is closer to the reference image. SSIM (Structural Similarity Index) measures perceptual similarity based on luminance, contrast, and structure. SSIM ranges from 0 to 1, where values closer to 1 indicate better structural similarity. In this experiment, the best parameter configuration was selected based on the highest SSIM.

## 4. Results Table

The table below reports the best spatial and frequency-domain configurations for every image.

| Image | Noise Type | Spatial Best Parameters | Spatial PSNR | Spatial SSIM | Frequency Best Parameters | Frequency PSNR | Frequency SSIM |
|---|---|---|---|---|---|---|---|
| camera | Gaussian noise | median, kernel=9x9 | 24.41 | 0.6469 | Gaussian LPF, cutoff=40 | 25.04 | 0.6879 |
| coins | Salt-and-pepper / impulse noise | median, kernel=3x3 | 28.07 | 0.8536 | Gaussian LPF, cutoff=40 | 23.38 | 0.5748 |
| text | Salt-and-pepper noise | median, kernel=3x3 | 32.01 | 0.9043 | Gaussian LPF, cutoff=20 | 25.59 | 0.6524 |
| rice | Salt-and-pepper noise | median, kernel=7x7 | 23.34 | 0.5659 | Gaussian LPF, cutoff=40 | 24.14 | 0.6168 |
| astronaut | Colored salt-and-pepper / impulse noise | median, kernel=3x3 | 30.32 | 0.9375 | Gaussian LPF, cutoff=40 | 22.33 | 0.6044 |

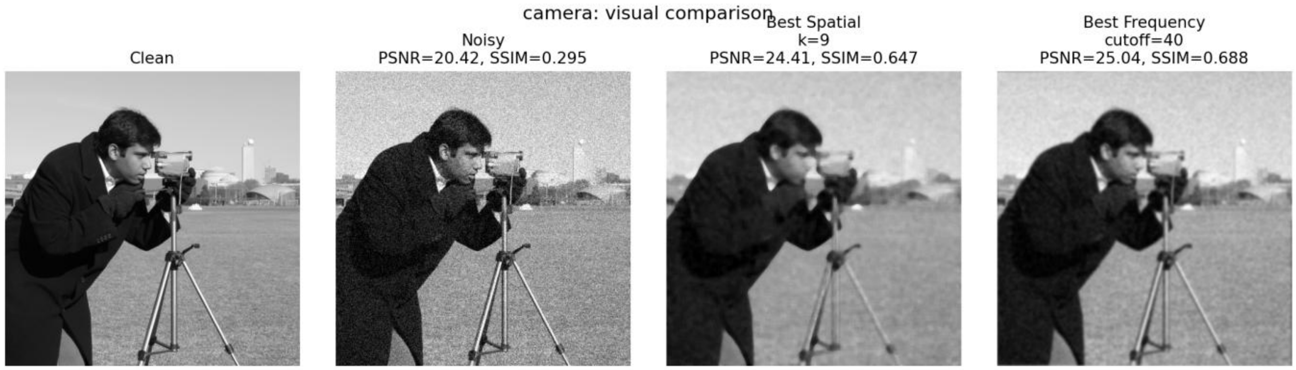
Average performance across the full dataset:

| Method | Average PSNR | Average SSIM |
|---|---|---|
| Spatial Median | 27.63 | 0.7816 |
| Frequency Gaussian LPF | 24.10 | 0.6273 |

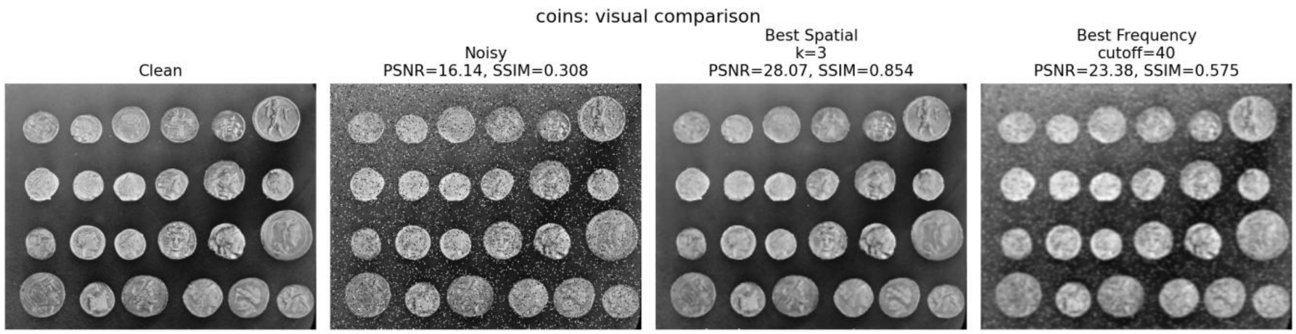
## 5. Visual Results

The following figures show clean reference, noisy input, best spatial output, and best frequency-domain output for each image.

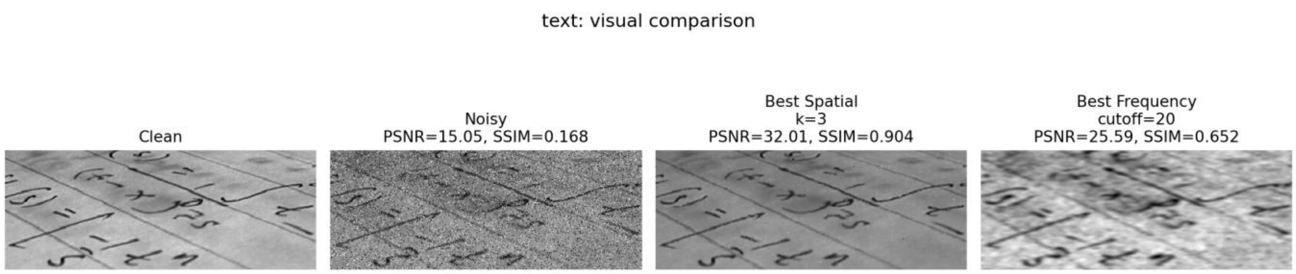
## Camera



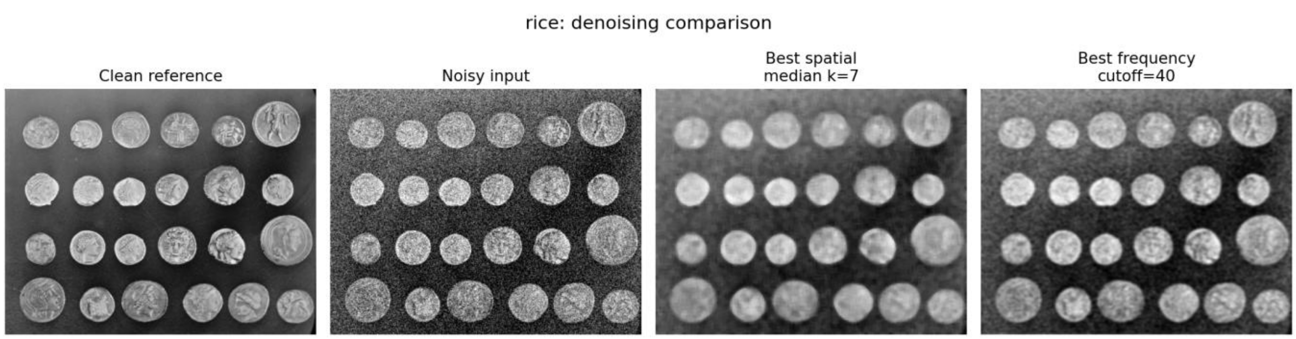

## Coins




## Text




## Rice



### Astronaut


## 6. Discussion

Based on the average results, the spatial median filter achieved an average PSNR of 27.63 and an average SSIM of 0.7816. The frequency-domain Gaussian low-pass filter achieved an average PSNR of 24.10 and an average SSIM of 0.6273. Median filtering performed better overall because the dataset contains several images with salt-and-pepper or impulse noise. The frequency-domain method was useful for smoothing, but its low-pass nature reduced high-frequency details such as edges and texture.

## 7. Conclusion

The assignment requirements were completed by implementing and comparing a spatial median filter and a Fourier-domain Gaussian low-pass filter. Multiple parameter settings were tested for both methods, and the best configuration was selected using SSIM. Overall, the spatial median filter provided the best denoising performance for this dataset.

## Appendix A: Full Parameter Tuning Table

| Image | Method | Parameters | PSNR | SSIM |
|---|---|---|---|---|
| camera | Noisy baseline | nan | 20.42 | 0.2951 |
| camera | Spatial Median | kernel=3x3 | 25.71 | 0.4939 |
| camera | Spatial Median | kernel=5x5 | 26.21 | 0.6040 |
| camera | Spatial Median | kernel=7x7 | 25.48 | 0.6442 |
| camera | Spatial Median | kernel=9x9 | 24.41 | 0.6469 |
| camera | Frequency Gaussian LPF | cutoff=10 | 20.54 | 0.5714 |
| camera | Frequency Gaussian LPF | cutoff=20 | 22.57 | 0.6268 |
| camera | Frequency Gaussian LPF | cutoff=40 | 25.04 | 0.6879 |
| camera | Frequency Gaussian LPF | cutoff=80 | 26.94 | 0.6403 |
| coins | Noisy baseline | nan | 16.14 | 0.3079 |
| coins | Spatial Median | kernel=3x3 | 28.07 | 0.8536 |
| coins | Spatial Median | kernel=5x5 | 26.17 | 0.7710 |
| coins | Spatial Median | kernel=7x7 | 25.12 | 0.7262 |
| coins | Spatial Median | kernel=9x9 | 24.30 | 0.6936 |
| coins | Frequency Gaussian LPF | cutoff=10 | 19.40 | 0.4628 |
| coins | Frequency Gaussian LPF | cutoff=20 | 21.61 | 0.5585 |
| coins | Frequency Gaussian LPF | cutoff=40 | 23.38 | 0.5748 |
| coins | Frequency Gaussian LPF | cutoff=80 | 22.50 | 0.4816 |
| text | Noisy baseline | nan | 15.05 | 0.1681 |
| text | Spatial Median | kernel=3x3 | 32.01 | 0.9043 |
| text | Spatial Median | kernel=5x5 | 28.51 | 0.8240 |
| text | Spatial Median | kernel=7x7 | 25.48 | 0.7166 |
| text | Spatial Median | kernel=9x9 | 23.76 | 0.6270 |
| text | Frequency Gaussian LPF | cutoff=10 | 23.71 | 0.5779 |
| text | Frequency Gaussian LPF | cutoff=20 | 25.59 | 0.6524 |
| text | Frequency Gaussian LPF | cutoff=40 | 25.33 | 0.5466 |
| text | Frequency Gaussian LPF | cutoff=80 | 21.26 | 0.3501 |
| rice | Noisy baseline | nan | 17.39 | 0.2941 |
| rice | Spatial Median | kernel=3x3 | 23.03 | 0.4602 |
| rice | Spatial Median | kernel=5x5 | 23.78 | 0.5393 |
| rice | Spatial Median | kernel=7x7 | 23.34 | 0.5659 |
| rice | Spatial Median | kernel=9x9 | 22.75 | 0.5644 |
| rice | Frequency Gaussian LPF | cutoff=10 | 19.69 | 0.4630 |
| rice | Frequency Gaussian LPF | cutoff=20 | 22.02 | 0.5731 |
| rice | Frequency Gaussian LPF | cutoff=40 | 24.14 | 0.6168 |
| rice | Frequency Gaussian LPF | cutoff=80 | 23.75 | 0.5054 |
| astronaut | Noisy baseline | nan | 14.56 | 0.2424 |
| astronaut | Spatial Median | kernel=3x3 | 30.32 | 0.9375 |
| astronaut | Spatial Median | kernel=5x5 | 27.59 | 0.8926 |
| astronaut | Spatial Median | kernel=7x7 | 25.71 | 0.8461 |

| | | | | |
|---|---|---|---|---|
| astronaut | Spatial Median | kernel=9x9 | 24.36 | 0.8019 |
| astronaut | Frequency Gaussian LPF | cutoff=10 | 17.13 | 0.4188 |
| astronaut | Frequency Gaussian LPF | cutoff=20 | 19.82 | 0.5283 |
| astronaut | Frequency Gaussian LPF | cutoff=40 | 22.33 | 0.6044 |
| astronaut | Frequency Gaussian LPF | cutoff=80 | 22.84 | 0.5150 |